Paleomagnetism and Age of Nonacho Group Sandstones and Associated Sparrow Dikes, District of Mackenzie[1]

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The Nonacho Group is a folded sequence of sedimentary rocks resting on a crystalline basement of probable Archean age. Both sedimentary and basement rocks are intruded by the Sparrow dikes. Fifty-four samples from 10 dikes have a stable magnetization with mean direction after a.f. cleaning of 131°, +51° ($N = 10$, $k = 50$, $\alpha_{95} = 7°$) and a corresponding pole at 12° N, 069° W. The presence of reversals and the approximate agreement between dikes and their baked contracts are taken to indicate that the dikes were probably magnetized at the time of intrusion. ^{40}Ar-^{39}Ar studies suggest that the probable age of intrusion was *about* 1700 m.y. The magnetization of the Nonacho sedimentary rocks collected at 12 sites (40 samples) have a mean direction after thermal cleaning of 148°, +57° ($N = 12$, $k = 11$, $\alpha_{95} = 14°$) without correction for geological dip. The corresponding pole is at 13° N, 086° W. The precision after correction for geological dip is low ($k = 3$), so the magnetization is predominantly post-folding. The thermal decay characteristics and the geometry of the directions indicate that the magnetization of the sedimentary rocks was acquired during cooling following a period of mild heating presumably owing to deep burial. This process of secondary magnetization apparently reached its culmination between *about* 1700 and 1600 m.y., and continued with diminishing intensity until *about* 1400 m.y. These results are used to modify the apparent polar wandering path for the Canadian Shield, so that the path now has a sharp bend (Hairpin 40) at about 1800 m.y. rather than at 1950 m.y. as previously postulated.

Le groupe Nonacho est formé d'une suite de roches sédimentaires plissées reposant sur un socle cristallin probablement d'âge Archéen. Les dikes de Sparrow font intrusions dans les roches du socle et les roches sédimentaires. Cinquante quatre échantillons provenant de 10 dikes ont un magnétisme stable avec une direction moyenne, après nettoyage a.f., de 131°, + 51° ($N = 10$, $K = 50$, $\alpha_{95} = 7°$), le pôle correspondant se trouvant au 12° N, 069° W. La présence d'inversions et la concordance approximative entre les dikes et leur zone de trempe indiqueraient que les dikes ont été magnétisés au moment de l'intrusion. Les études ^{40}Ar-^{39}Ar suggèrent un âge possible d'*environ* 1700 millions d'années pour l'intrusion. Le magnétisme des roches sédimentaires du Nonacho, échantillonnées en 12 endroits (40 échantillons), a une direction moyenne, après nettoyage thermique, de 148°, +57° ($N = 12$, $K = 11$, $\alpha_{95} = 14°$) sans correction pour le pendage géologique. Le pôle correspondant se situe au 13° N, 086° W. La précision après la correction pour le pendage géologique est mauvaise ($K = 3$) aussi la magnétisation est surtout postérieure au plissement. Les caractéristiques de la dégradation thermique et la géométrie des directions indiquent que le magnétisme des roches sédimentaires fut acquis durant le refroidissement qui a suivi une période de léger réchauffement du sans doute à un enfouissement profond. Ce processus de magnétisation secondaire atteignit apparamment son point culminant entre environ 1700 et 1600 m.a. et s'est poursuivi en diminuant d'intensité jusqu'à *environ* 1400 m.a. Ces résultats sont utilisés pour modifier la trajectoire du déplacement apparent du pôle pour le Bouclier Canadien de telle sorte que la trajectoire a maintenant une courbe prononcée (Hairpin 40) vers environ 1800 m.a. au lieu de 1950 m.a. comme postulé auparavant.
[Traduit par le journal]

Introduction

This work forms part of a paleomagnetic study of late Aphebian-Paleohelikian (middle Proterozoic, approximately 1800 to 1350 m.y.) rocks of the western Canadian Shield. The purpose of this program is to determine the chronology of paleomagnetic variations, and then to use this chronology as a stratigraphic tool for geological correlation, and as a tectonic tool to determine the nature of con-

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tinental drift in the Precambrian. The first results of this program have already been published (Irving, Park, and McGlynn 1972; Irving, Donaldson, and Park 1972; Park *et al.* 1973) and reviewed by Donaldson *et al.* (1973).

The present study had three objectives. The first was to determine the directions of magnetization and corresponding poles for the Nonacho Group and associated basic dikes (referred to as the Sparrow dikes, from Sparrow Bay where they are well displayed). The second was to estimate the age of the dikes by the ^{40}Ar–^{39}Ar method using step-wise heating. The third was to ascertain when, during the history of the region, the magnetization was acquired. We have assembled much information, but have been only partly successful in achieving these objectives because the long and complex history of the region has obscured the record. The problems are typical of those encountered in magnetic and age studies of rocks that have undergone burial to many thousands of meters (and heated to 100 or 200 °C), uplift, folding, and faulting. In the field such rocks appear essentially unaltered. In thin section they are mildly altered, but it is difficult to determine whether this is owing to deuteric (in igneous) and diagenetic (in sediments) changes that occurred soon after formation, or to mild metamorphism during burial long afterward. The changes that this alteration signifies may be sufficient to reset the magnetization and to disturb the accumulation of radioactive nucleides. If the alteration occurred penecontemporaneously then the record may be directly interpreted, but if it occurred long after, the record may be updated perhaps by hundreds of millions of years. Thus, in the present study, although we have been able to achieve a reconstruction of events consistent with much diverse data, the mild alteration that these rocks have undergone make our conclusions necessarily tentative.

Geology

The Nonacho Group (Fig. 1, general sampling area marked in Fig. 5) is a conformable sequence of lithic sandstones, polymictic conglomerates, and shales, resting on a granitic basement (McGlynn 1970; 1971). The most abundant clasts in the conglomerate are of granitic lithologies, many of which can be recognized in the basement, and white or gray metaquartzites, Vein quartz, basic volcanic rocks, intermediate porphyritic volcanic rocks, and metasediments, which cannot be matched in the basement, are also found. The dominant rock type of the Nonacho sequence is fine- to coarse-grained lithic sandstone, which may be cross-bedded, massive, or laminated, and which is made up of rock fragments, quartz, and feldspar, with a quartz and occasionally carbonate cement. Interbedded shales and sandstones or shales dominate locally.

The Nonacho rocks are folded about steeply dipping northeast-trending axial planes. The folding is open in sandstones and conglomerates, but dips are steep where shales become abundant. The folds plunge gently. Faults are concentrated along the margins of the basin (especially in the west) and about basement highs within the basin. These faults are parallel to the basin, and movement along them affects both basement and Nonacho rocks. Some movement probably occurred prior to Nonacho sedimentation, because clasts of mylonite occur in the conglomerates. Faulting continued during and after sedimentation, and folding of the sediments may be related to some of these later fault movements.

Where faulting was intense the relations of the Nonacho sediments to the basement are obscured and the unconformity is often destroyed. However, all facies have been observed to rest on basement along both margins of the basin. The basement rocks include mylonite, crushed granitic rocks, flaser gneisses, migmatites, and massive and gneissic granitic rocks.

The basement and Nonacho rocks are cut by the Sparrow dikes. These are narrow (generally 1–6 m wide), vertical, basic dikes, which occur in clusters around the margins of the basin or about basement highs within the basin. The strike of the dikes is northerly or slightly west of north. They cut both basement and sedimentary rocks, but are rare in the central parts of the basin. The dikes are not displaced by faults, but are broken by fractures (although the rock itself is not internally fractured), suggesting that they were emplaced at about the time of the last minor movements on the faults marginal to the basin. The dikes

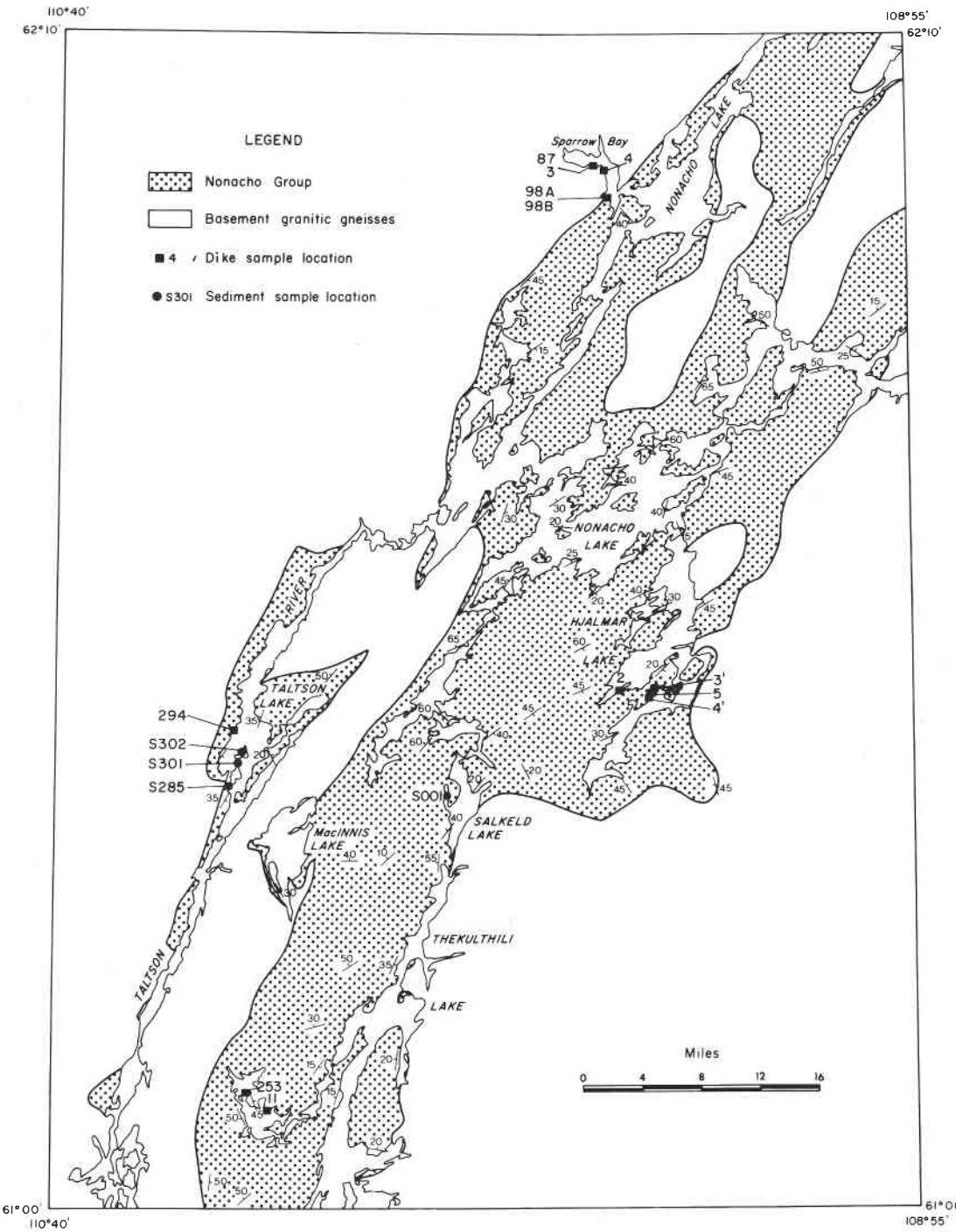


FIG. 1. Geology of the Nonacho Lake area and the location of sampling sites. The site numbers are laboratory numbers and have no relation to the geology.

are commonly amygdaloidal, often contain inclusions of the wall-rocks, and are generally fine grained. They consist of laths of basic plagioclase with pyroxene, minor quartz, and commonly sphene. The dikes of the western margin (Sparrow Bay and Taltson Lake) are fresh, with only minor chlorite and amphibole developed. The dikes from Hjalmar Lake are more altered, pyroxene is partly changed to amphibole and chlorite, and plagioclase to white mica and carbonate.

Radiometric Evidence of Age

The age of the basement rocks is not well defined. Biotites from the granitic rocks in the vicinity of the Nonacho basin gave K–Ar ages ranging from 1850 m.y. to 1735 m.y., muscovites 2175 m.y. to 1745 m.y. (Wanless *et al.* 1963), and one hornblende 2240 m.y. (Burwash and Baadsgaard 1962). In the more general region south of Great Slave Lake and north of latitude 60°, K–Ar ages of micas range from 2555 m.y. to 1735 m.y., with most between 1900 and 1735 m.y. It has been suggested that the granitic basement rocks are in fact Archean, and that the younger K–Ar ages signify a later event consisting largely of faulting and uplift contemporaneous with, and perhaps related to, the Hudsonian orogeny elsewhere in the Churchill Structural Province (McGlynn 1970).

Whole-rock K–Ar ages from the Sparrow dikes have been obtained by the Geological Survey of Canada as follows: site 4, 1480 m.y., site 98B, 1390 m.y., and a site just east of Thekulthili Lake, 1550 m.y. These are probably minimum ages.

The new data are given in Table 1 and Fig. 2. The accuracy of a $^{40}Ar/^{39}Ar$ ratio for a given temperature is about 3%, based on the uncertainties in the determination of the $^{40}Ar/^{39}Ar$ ratio of the standard hornblende NL-25-II with an age of 2700 m.y. (G. N. Hanson, unpublished data). The precision for any temperature fraction containing more than a few per cent of the total ^{40}Ar is about 1%. The rock chips (100 to 200 mg) along with monitor NL-25-II hornblende were irradiated in a fast neutron flux at Brookhaven National Laboratory. Corrections for Ca-derived ^{39}Ar and ^{36}Ar were made using values for pure calcium salts (M. A. Hussein 1972, oral communication). Each step was for one hour of heating. The temperatures were checked using a pyrometer and are good to only 50 °C, as there was a 50–100 °C difference between the crucible and the top of the sample.

Sample 87 (basalt) is altered and contains amoeba-like pods of chlorite, carbonate, and quartz. There is some sericitic alteration of the plagioclase, but it does not obliterate the plagioclase, and albite twins can still be seen. In general the clinopyroxene is quite fresh. Sample 98A (also a basalt) is much less altered. There is only minor carbonate, chlorite occurs as a minor alteration of pyroxene, and the plagioclase is only slightly sericitized. Samples 253 and 11 (microgabbros) have interstitial micrographic intergrowths of quartz, plagioclase, and minor amphiboles and biotite. Both are slightly altered, with pods of carbonate, chlorite, and quartz, similar to sample 87. The plagioclase is sericitized, but not enough to obscure the grains.

The least altered sample (98A) gives the simplest pattern, indicating an age of intrusion of about 1700 m.y. in the temperature range 900–1100 °C (Fig. 2). The altered samples give more complicated results, the ages generally increasing from low to high temperature. Samples 253 and 11 increase to an age of about 1700 m.y. at 1100 °C, and 87 increases to almost 2000 m.y. at 1200 °C. The latter age may be too high because altered dike rocks can inherit radiogenic argon and give erroneously high apparent ages. Such an instance has been found in Minnesota where an olivine diabase dike that has undergone prehnite-pumpellyite-grade metamorphism, and which cuts 2700 m.y. basement rocks, gave an anomalously high age of 3825 m.y., and clearly contains excess radiogenic argon (Hanson and Malhotra 1971).

The radiometric evidence does not provide a clear-cut age for the Sparrow dikes, presumably owing to later thermal history. The most reliable radiometric estimate is that of *about* 1700 m.y. from the least altered dike (sample 98A).

Paleomagnetism of Nonacho Sandstones

Core samples were oriented by solar and magnetic methods. A specimen from each site was thermally demagnetized in an electrical

TABLE 1. Analytical data for ^{40}Ar-^{39}Ar dating

Sample	Fraction T (°C)	% Air ^{40}Ar in Fraction	% of Total ^{40}Ar Released	Age[1] (m.y.)
98A	700	35	1.1	1165
	800	7.0	9.8	1605
	900	0.95	25.9	1710
	1000	0.6	24.8	1645
	1100	0.7	30.6	1635
	1200	2.0	7.4	1510
	1600	7.0	0.4	1595
	weighted mean[2]			1630
87	700	14.7	3.9	1705
	800	5.4	5.6	1685
	900	1.0	17.2	1575
	1000	0.4	30.0	1650
	1100	0.9	21.6	1810
	1200	1.6	19.1	1965
	1600	4.6	2.5	1905
	weighted mean[2]			1750
253	700	47.4	1.3	835
	800	10.3	6.8	1245
	900	1.2	22.3	1495
	1000	0.8	20.7	1520
	1100	1.0	21.4	1595
	1200	1.5	23.4	1705
	1600	2.1	4.2	1720
	weighted mean[2]			1560
11	700	35.9	2.9	1130
	800	5.8	9.4	1435
	900	1.6	20.0	1510
	1000	1.5	22.5	1410
	1100	1.3	19.1	1605
	1200	1.7	22.1	1775
	1600	4.2	4.1	1555
	weighted mean[2]			1550

[1]^{40}K = 0.0119 Atom % of K; $\lambda e = 0.584 \times 10^{-10}$ yr^{-1}; $\lambda\beta = 4.72 \times 10^{-10}$ yr^{-1}.
[2]These are the means of the values in the right-hand column weighted according to the amount of ^{40}Ar released; they would correspond to whole-rock K–Ar ages from these samples.

furnace (Roy *et al.* 1972). Only one site (001, a banded sandstone) gave square-shouldered curves with blocking temperatures in the range 650 to 700 °C (Fig. 3). All other specimens are of fine-grained maroon sandstone and show a lower and wider range of blocking temperatures from below 100 to 650 °C; the mean ratio (11 sites) of intensities before and after thermal demagnetization at 500 °C is 0.53.

The opaque minerals occur as subrounded grains (over 100 microns with corroded margins) that may be detrital, as abundant very fine particles commonly distributed along boundaries of detrital silicate grains, and as grains within the lithic fragments. The relative proportions of each mode vary, but all were present at all collecting sites. Most of the opaque minerals are hematite and altered ilmenite (leucoxene). There are also rare grains of rutile and sulfides. Red staining, presumably hematite, is abundant in all samples. Magnetite was not identified. The only visible carrier of the observed remanence is hematite, although typical blocking temperatures in the maroon sandstone are many hundreds of degrees lower than the Curie temperature of hematite (675 °C). Such distributed blocking temperatures are typical of viscous partial thermo-remanent magnetization (VPTRM), that is, of magnetization acquired over long periods of time by heating to temperatures (followed by cooling) lower than the Curie temperature (Chamalaun 1964; Irving and Opdyke 1965), but this is not to say that they may not be pro-

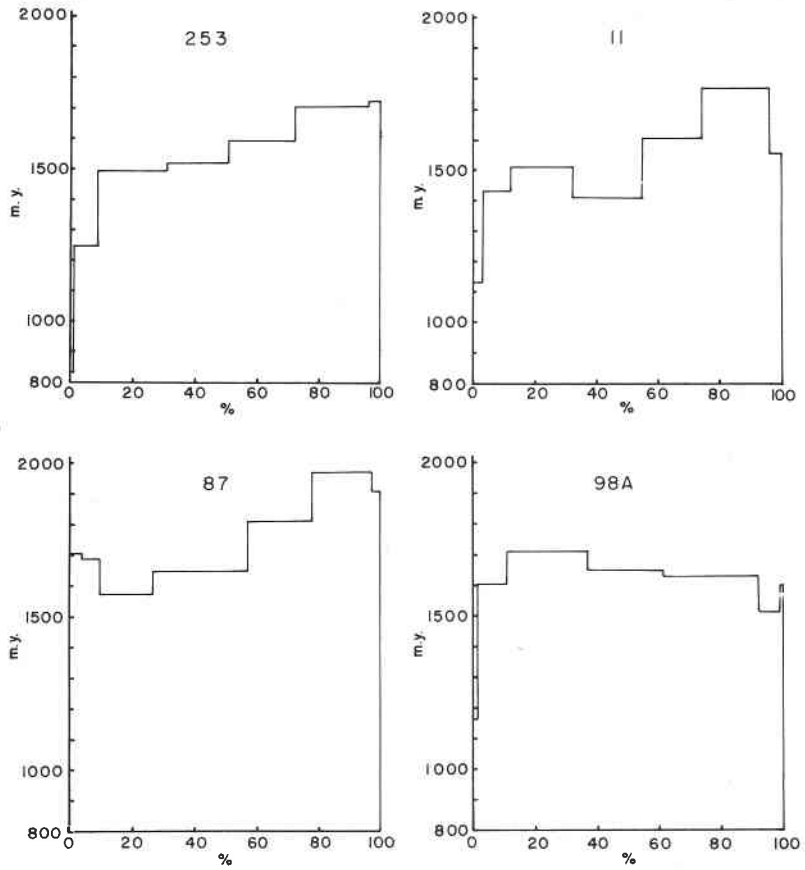


FIG. 2. ^{40}Ar–^{39}Ar dating. Apparent ages of different temperature fractions, respectively 700, 800, 900, 1000, 1100, 1200, and 1600 °C.

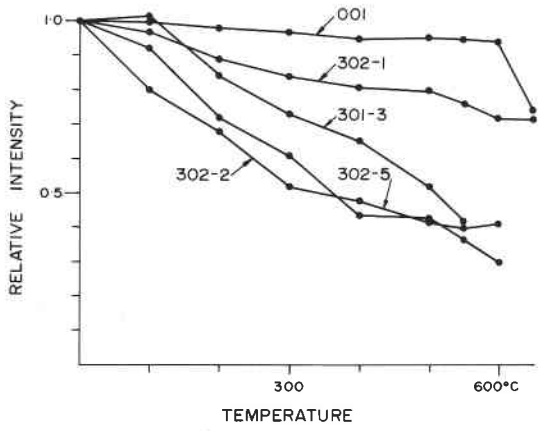


FIG. 3. Thermal demagnetization of Nonacho sandstones. Relative intensity as function of temperature for five specimens showing range of behavior observed. The curves terminate in range 550 to 650 °C. At high temperatures the directions become random.

duced by other processes. VPTRM's with all the main characteristics observed in red beds can be produced in the laboratory by heating to 300 °C for 3 weeks (Roy *et al.* 1967), the build-up being proportional to the logarithm of time, as Neel (1949) predicted. Over geological periods of time, therefore, stable VPTRM's may be expected to occur in rocks, subjected to temperatures of about 100 to 200 °C, temperatures to which the Nonacho sediments have probably been subjected (see below).

The direction changes during heating in the range 100 to 550 °C are either negligible (site 001) or are systematically toward the south and southeast, away from the direction of the present geomagnetic field by amounts varying from 10° to 40°. The mean site directions after thermal cleaning are given in Table 2 and Fig. 4. The directions, when restored so that bedding planes are horizontal, are widely

TABLE 2. Directions of magnetization of Nonacho sedimentary rocks

Site	n	M_n	T(°C)	D,I	k	Pole
001	6	1179.0	500	084,+33	36	18, 023
285	2	4.8	550	128,+56	28	17, 069
301(1)	4	5.6	500	199,+49	96	03, 127
301(2)	3	4.3	500	193,+61	38	14, 120
301(3)	3	2.3	550	219,+42	10	01, 145
302(1)	2	6.0	550	128,+46	27	08, 065
302(2)	4	4.8	550	175,+62	5	14, 106
302(3)	6	5.5	550	132,+48	65	08, 069
302(4)	3	6.9	550	139,+43	190	03, 074
302(5)	3	5.1	550	133,+59	43	18, 075
302(6)	3	5.4	550	148,+56	112	12, 085
302(7)	1	5.9	550	136,+55	—	13, 075

NOTE: n is the number of cores (two specimens measured from each). M_n is the initial (n.r.m.) intensity in cgs units $cm^{-3} \times 10^{-6}$. T is the cleaning temperature. D, I are the declination and inclination in degrees, and k the precision (Fisher 1953) calculated giving unit weight to cores (n). The corresponding poles are specified by latitude (N) and longitude (W).

scattered ($k = 3$). Without this correction the directions are better grouped ($k = 11$). Therefore most of the remanence was acquired after folding, but there is still substantial residual dispersion. This could be caused either by the retention of a small prefolding component, by changes in field direction during the time the rocks were magnetized, or by anisotropy. We shall return to this problem later, and deal here only with anisotropy.

The effect of anisotropy on the remanence directions has been studied using the method of Irving and Park (1973). A specimen from each site has been reoriented into its position at the outcrop and a field applied by an electromagnet in a direction parallel to the observed stable remanence. The isothermal remanence (i.r.m.) acquired is measured and then demagnetized to determine the effect of anisotropy in the different coercive force fractions. The dispersion of directions is a measure of the individual departures from the applied field owing to anisotropy. Departures of the mean from the applied field are a measure of any systematic effect owing to anisotropy. The dispersion decreases after demagnetization, so the high coercive force components are less affected by anisotropy than the low coercive force components (Table 3). This is now known to be a common effect (Park 1973) and was first discovered by Kern (1961). The mean directions are 3° more steeply inclined than the applied field, as if the i.r.m. is deflected toward the bedding planes, but the effect is small, and not statistically significant. Therefore it is unlikely that the residual dispersion referred to above is caused by anisotropy.

Paleomagnetism of Sparrow Dikes

The results are given in Table 4 and plotted in Figs. 4 and 5. Ten dikes respond well to magnetic cleaning, whereas two (3′ and 5) do not. The latter have very low coercivities and their magnetization is probably of recent origin (see footnote to Table 5). During cleaning the directions in most dikes move away from the present field (Fig. 5), and there is usually a decrease in dispersion. In dike 2 there is an increase, probably owing to instrumental noise since the intensities are unusually low (Table 4) for an igneous rock and fall to values of about 2×10^{-6} (Table 4, Fig. 5). In dike 11 (Fig. 6) the directions do not trend away from the present field but at first toward it, and then beyond into the southeast quadrant, as if the n.r.m. is the sum of two approximately antiparallel components (such as have been found by Roy and Robertson (1969) and Roy and Park (1972)), the NW negative component being softer magnetically. If the two unstable dikes (3′ and 5) are excluded, the overall grouping is good (Table 5, Fig. 4). The mean direction is not significantly different from the mean direction of the sedimentary rocks.

Only one of the ten stable Sparrow dikes (11 and 253) is wide (about 30 m) and coarse grained. It contains numerous large (500 microns) roughly rectangular grains of magnetite with abundant intergrowths of pinkish ilmenite. Surfaces have a granulated ap-

TABLE 3. Anisotropy of remanent magnetization

	n	Applied field D,I	Observed remanence D,I	k	α_{95}
Nonacho sediments i.r.m.	12	148,+57	150,+60	67	10
Nonacho sediments 500 OE A.F.	12	148,+57	148,+60	126	7
Sparrow dikes i.r.m.	6	131,+51	129,+47	98	7
Sparrow dikes 250-500 Oe A.F.	6	131,+51	129,+51	606	3
Baked contacts i.r.m.	6	131,+51	143,+55	53	9
Baked contacts 200-500 Oe A.F.	6	131,+51	143,+53	43	10

Symbols as for Tables 2 and 5. The i.r.m. was imposed in a field of 20 000 Oe in the mean direction of the stable component of n.r.m. (Table 5). For the sediments one specimen from each site was used. One specimen from each of the six baked contacts and their associated dikes was used.

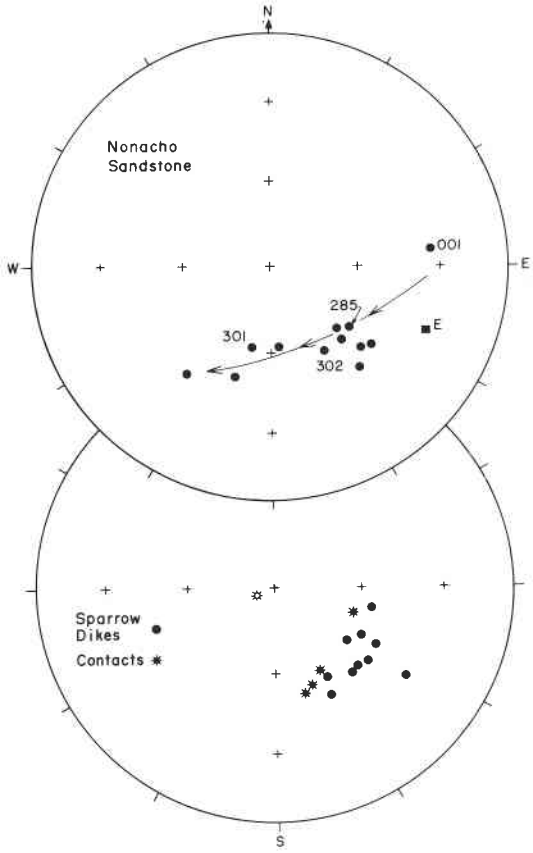


FIG. 4. Mean directions of magnetization in Nonacho Group sandstones (12 sites), Sparrow dikes (10 samples) and their baked contacts (5 samples). The primitive is the present horizontal. The square marked E is the direction in the nearby Et-Then Group (Irving, Park, and McGlynn 1972). All directions are down except one contact marked with an open symbol.

pearance, indicating a moderate degree of alteration. Abundant skeletal pyrite is present. The directional precision is high (Table 5).

The other nine stable dikes are narrow (1–6 microns) and typically contain large skeletal titanomagnetites (100–500 microns) together with roughly rectangular magnetite. Both minerals commonly show incipient ilmenite intergrowths. In addition, abundant very small disseminated titanomagnetite grains (5 microns and less) are present. Clearly their stability is due to the latter feature or to intergrowths of very fine scale.

Of the two unstable bodies, dike 3′ has blocky titamagnetite, typically 5–20 microns, devoid of intergrowths, and therefore of large effective grain-size. Dike 5 is very altered, with no identifiable magnetic minerals but with abundant pyrite, features that are presumably responsible for the unsatisfactory results.

The effect of anisotropy on the remanence of the Sparrow dikes is summarized in Table 3. The agreement with the applied field after magnetic cleaning (the fields used are those determined as the optimum cleaning fields for n.r.m.) is good, but the dispersions ($k = 98$ before and $k = 606$ after magnetic cleaning), although small compared with those of n.r.m. (Table 5), are appreciable, and much greater than corresponding values ($k = 2000$) determined for much wider diabase dikes of the Mackenzie swarm (Irving and Park 1973). Therefore in these thin dikes a small, but paleomagnetically insignificant remanence anisotropy is present, presumably owing to laminar flow confined within a narrow duct.

Paleomagnetism of Baked Rocks

The magnetization of baked rocks within 30 cm of contacts (three sites in sedimentary rocks and three in basement gneisses) show several interesting features. (*1*) Their intensities are higher than similar material, collected more than 100 m from contacts (compare

TABLE 4. Directions of magnetization of Sparrow dikes and their baked contacts

Site	n	M_n	a.f.	D,I	k	Pole
002 dike	7	24	500	306, −34	100	01, 059
002 baked sediment	3	203	500	334, −56	290	10, 009
003 dike	5	1452	300	138, +50	276	09, 074
003 baked gneiss	2	84	500	160, +53	91	07, 093
003′ dike	5	597	300	019, +85	11	71, 098
004 dike	4	1921	300	130, +49	632	10, 068
004 baked gneiss	2	11	200	109, +62	32	29, 057
004′ dike	5	1926	400	333, −47	2018	03, 086
005 dike	5	555	300	348, +83	20	75, 120
005 baked sediment	4	8	500	246, −83	282	64, 081
011 dike	5	5250	250	121, +56	467	19, 064
011 baked sediment	6	145	200	074, +66	4	48, 037
087 dike	9	107	400	136, +50	275	09, 072
087 baked gneiss	2	10	400	165, +51	801	05, 096
253 dike	5	4179	250	129, +59	1197	19, 072
294 dike	4	1714	150	152, +54	527	09, 087
098A dike	5	2570	300	121, +50	195	14, 060
098B dike	5	2647	300	103, +56	37	26, 049

Symbols and conventions as for Table 2.

TABLE 5. Synopsis of paleomagnetic results

	N	D,I	k	α_{95}	Pole
1. Nonacho Group sandstone directions *in situ*	12	148, +57	11	14	13, 085
2. Nonacho Group sandstones poles	12	—	06	19	13, 086
3. Sparrow dike directions	10	131, +51	50	07	12, 069
4. Sparrow dike poles	10	—	35	08	12, 069
5. Sparrow dike baked contacts only	4	150, +57	33	16	13, 086
6. Sparrow dikes and baked contacts	10	134, +54	54	07	13, 073

NOTES: N is the number of bodies each given unit weight. D,I are the declination and inclination of the mean direction, k the precision, and α_{95} the radius of the circle of confidence with $P = 0.05$ (Fisher 1953). The poles (lat. N and long. W) in entries 1, 3, 5, and 6 are calculated from the mean directions and those in entries 2 and 4 are calculated as an average of site poles listed in Tables 2 and 4. Entry 1 and 2 are calculated *without* correction for geological dip. Entry 3 is for 10 reliable dike directions (8 normal and 2 reversed); results from dikes 3′ and 5 are excluded because of their low precision, their low coercivities (m.d.f. or n.r.m. are 57 and 34 Oe respectively), and because their directions are near the present geomagnetic field direction so that their magnetization is very probably of recent origin. Entry 5 is an average of four baked contacts; contact at dike 5 is excluded because it is widely divergent from the main group, and 11 is excluded because of its low coercivity (m.d.f. of n.r.m. is 40 Oe) and low precision (Table 4).

Tables 2 and 4). (*2*) The rocks baked by dike 11 are very unstable (see footnote Table 5). (*3*) The baked contact of the unstable dike 5 is very stable, the directions being well grouped and about 50° away from the nearest directions observed in Sparrow dikes (Fig. 4). Demagnetization studies show that only one component is present. Hence this may represent an intermediate direction during field reversal. (*4*) The remaining four baked contacts (2, 3, 4, 87) have directions *approximately* similar to those in the adjacent dikes, but displaced from them commonly by about 20°; and the displacements are systematic, the mean direction in these contacts having a declination 15° greater than the dikes (see for example Fig. 5, sites 2 and 87). The anisotropy of i.r.m. is similarly deflected from the applied field (Table 3) by about 12°, although the scatter is such that the effect appears not to be significant. We cannot, therefore, be sure that the differences between baked contacts and their associated dikes are owing to anisotropy, but it could be the cause.

The mean of the coercivities (median destructive field) of n.r.m. of baked contacts is 320 Oe, and that of dikes 144 Oe. The opaque minerals in samples of baked basement gneiss are magnetite, pyrite, and minor goethite. They occur either as rather large blocky grains adjacent to mafic silicate minerals, or as small grains disseminated inside mafic minerals, commonly along cleavage.

In baked sedimentary rocks the opaque

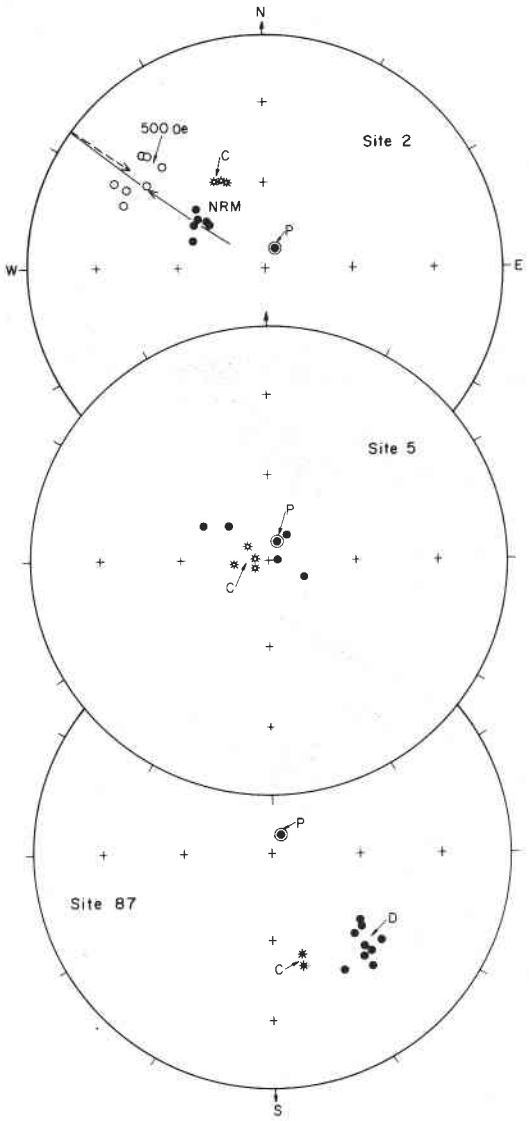


FIG. 5. Directions in dikes and baked contacts compared. Above, a typical migration of directions showing removal of component along present field during cleaning. In the middle, unstable dike sample directions around present field, and stable contact sediments with upward directions, are shown. Below, a typical divergence of contact and dike rocks is shown.

minerals are variable. Red staining is present in all samples. At two sites they consist of pyrite (finely disseminated or as small cubes) and altered ilmenite. At two others, finely disseminated hematite and altered ilmenite occur. At a fourth site, the opaques consisted of magnetite and minor hematite.

Discussion and Conclusions

1. *Sparrow dikes.* The remanence directions of the Sparrow dikes are well-grouped ($k = 50$) and reversals occur (Fig. 4). The intensities of rocks baked by them are enhanced as expected if the remanence is primary. Most dike directions do not differ strongly from the direction in many of the country rocks, but three do differ (2, 4, 5). We argue therefore that the occurrence of reversals, and the enhanced intensities of baked rock, indicate that the remanence of the dikes probably originated at the time they were intruded.

When were the Sparrow dikes intruded? If the above argument is correct, then the good agreement between dikes indicates only one magmatic episode. The minimum age is about 1400 m.y., as given by the whole-rock K-Ar ages. The ^{40}Ar-^{39}Ar age from the least-altered dike sample is about 1700 m.y. Other more altered samples yield apparent ages of 1700, 1800, and 2000 m.y. We suggest that the most probable age of the Sparrow dikes is that given by the least altered sample, namely 1700 m.y. The Nonacho strata rest on basement that is considered to be Archean (McGlynn 1970), and are cut by the Sparrow dikes. It follows therefore that the sediments were deposited between about 2500 and 1700 m.y.

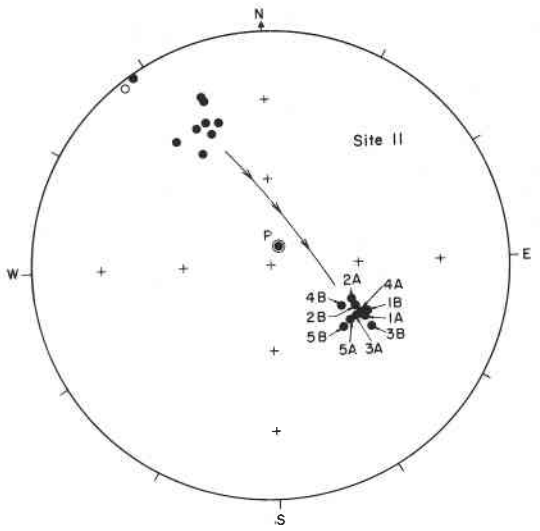


FIG. 6. Magnetic cleaning at site 11 showing removal of components not aligned along present field *P*.

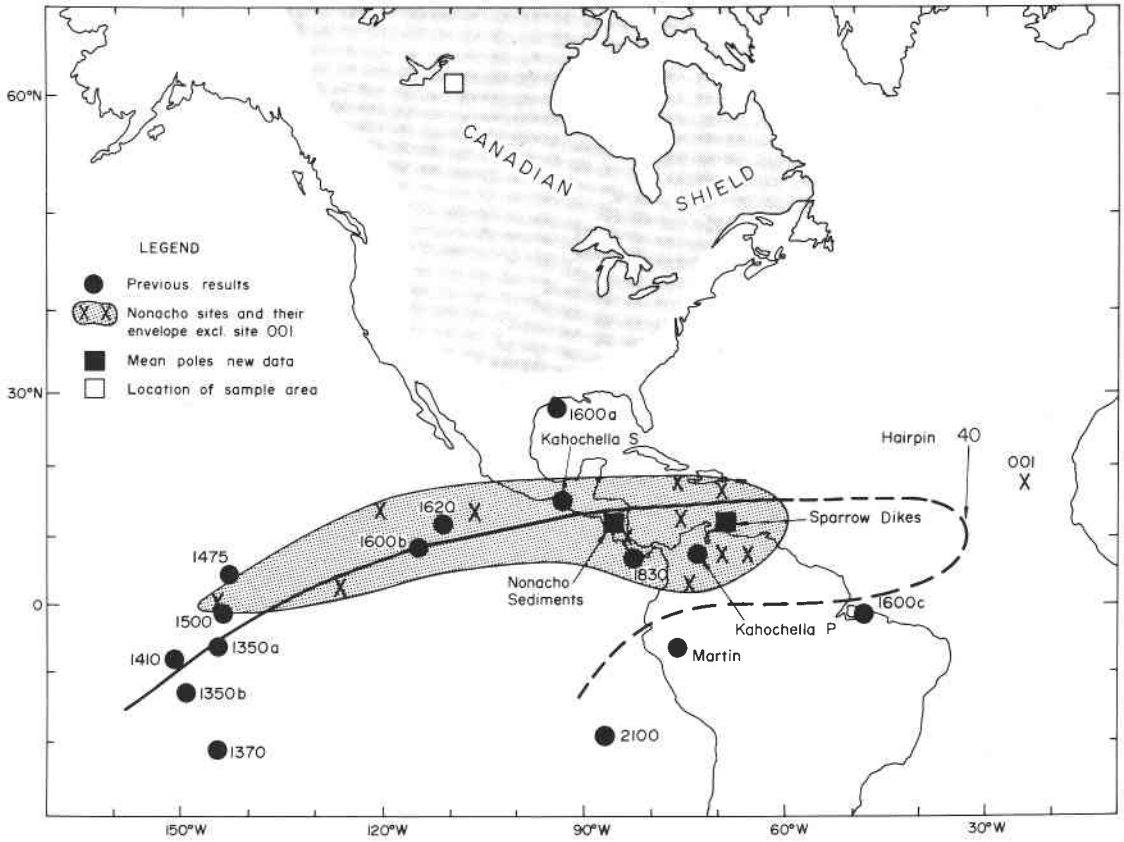


FIG. 7. Comparison of Laurentian poles in the interval 2100 to 1300 m.y. The results given in this paper and the recent result from the Great Slave Group are labelled. Earlier results are as follows with their estimated ages: Belt Series (1350*b*), Sibley Group (1370), Sherman Granite (1410), Michikamau Anorthosite (1500), rocks of St. Francois Mts. (1350*a*), Croker Island Complex (1475), Western Channel Diabase (1600*b*), Gunflint Formation 1600*a*), Dubawnt Group (1830), Et-Then Group (1600*c*). Assigned ages may be subject to considerable uncertainty.

2. *Apparent polar wander 2100–1300 m.y.* Paleomagnetic poles for the Canadian Shield are plotted on Fig. 7 and indexed by the assigned ages of Stewart and Irving (1973). These ages are not to be regarded as absolute ages. This notation is simply an attempt to summarize the existing data in a form that is easy to comprehend. Substantial errors, documented in Stewart and Irving (1973) are associated with these estimates; for example pole 1600*c* from the Et-Then Group is subject to an uncertainty of ±200 m.y. (Irving, Park, and McGlynn 1972).

The mid-Proterozoic poles form a trend from the Caribbean region westward to the east-central Pacific. The elapse time appears to range from *about* 1800 m.y. to about 1350 m.y. The pole for the Sparrow dikes agrees approximately with the results from the Dubawnt Group (1830; Park *et al.* (1973)), and with the pole from the red beds of the Kahochella Group of the Great Slave Supergroup (McMurray *et al.* 1973). The latter pole (Kahochella *P*) is thought by these authors to correspond to the primary magnetization, and a second pole (Kahochella *S*) to a secondary magnetization of the Kahochella Group (see below).

The relationship of the Et-Then pole (1600*c*) is mysterious. The Et-Then overlies unconformably the Great Slave Supergroup. Its pole therefore should lie to the west of

Kahochella *P*, rather than to the east where it is found. Bearing in mind the uncertainty in the age of the Et-Then pole, and that all poles have statistical errors of about 10°, the results can be reconciled if it is assumed that the pole moved in a tight loop, as shown by the broken line in Fig. 7. The time order of poles now becomes: Nipissing diabase (2100), Martin Formation (Evans and Bingham 1973), Dubawnt (1830), Kahochella *P*, Et-Then group (1600*c*), Sparrow dikes, Nonacho Group (see below), Kahochella *S*. This polar loop has been referred to as Hairpin 40 with an estimated age of 1950 m.y., and identified as the polar signature of the Hudsonian orogeny of Stockwell (1961) (Donaldson *et al.* 1973; Irving and Park 1972). The reconstruction of Fig. 7 implies that the age of Hairpin 40 is *about* 1800 m.y., somewhat younger than previously estimated, and closer to Stockwell's estimate of the age (1800 m.y.) of the end of the Hudsonian orogeny (Stockwell 1972).

3. *Nonacho sandstones.* The magnetization of the maroon Nonacho sandstone is predominantly secondary. The individual site poles extend along the polar path for the middle Proterozoic (Fig. 7). If the above argument is correct the age of this segment of the path is *about* 1700 m.y. to *about* 1400 m.y. This process of magnetization apparently extended over several hundred million years, beginning at about the time of intrusion of the Sparrow dikes and continued until 1400 m.y., the *apparent* date of the 'youngest' Nonacho site pole (Fig. 7). It is notable that the youngest whole-rock K–Ar age from the Sparrow dikes is also about 1400 m.y. Thus the blocking of the remanence and the retention of radiogenic argon in all its blocking intervals may occur at *about* the same time.

What is the reason for the 'young tails' of site poles for the maroon sandstones, and of whole-rock K–Ar dates in the associated dikes? The effects may be a response to temperature changes following uplift, or to cryptic chemical alteration, or to a combination of these. The effects of chemical alteration are very difficult to determine, because although the rocks are mildly altered the time of alteration is unknown. The effects of temperature can however be estimated.

The remanence has distributed blocking temperatures characteristic of viscous PTRM (see above). It is therefore possible that the remanence was acquired during uplift and cooling following deep burial. Neel's theory of magnetic viscosity (Neel 1949) states that the build-up of remanence is proportional to the logarithm of time

$$T_1 \ln C\,\tau_1 = T_2 \ln C\,\tau_2$$

where T_1 and T_2 are the temperatures corresponding to relaxation times τ_1 and τ_2, and C is a constant equal to $10^{10}\ s^{-1}$. The left-hand side of this equation corresponds to the initial build-up of remanence, so that T_1 is the temperature during burial and τ_1 the duration of burial. We assume a duration of 10^8 years from Fig. 7. The right-hand side of the equation corresponds to the destruction of this remanence in the laboratory, so that T_2 is the characteristic blocking temperature (which we may take as 500 °C from Fig. 3) and τ_2 is the time (about 3 min.) at which the specimen is held at that temperature during thermal demagnetization. Given these values T_1 works out to be about 200 °C, which is roughly the temperature below which argon is retained (Moorbath 1967). It is possible therefore that both the magnetization of the maroon sandstones and the 'young tail' of whole-rock K–Ar ages are the result of mild heating during burial. This is not to say that chemical changes may also be responsible wholly or in part.

From a study of the directional variation observed between 550 and 600 °C in red beds of the Kahochella Group, McMurray *et al.* (1973) show that a secondary component is present. This secondary component gives the Kahochella *S* pole (Fig. 7). Kahochella *S* agrees very closely in position and polarity with the mean Nonatcho sediments, whose magnetization we have shown to be secondary. The two localities are about 100 km apart. This agreement suggests that there was a rather widespread episode of remagnetization in this region during the middle Proterozoic. If the polar reconstruction of Fig. 7 is correct then the culmination of this remagnetization process (as represented by the mean Nonacho and Kahochella *S* poles) occurred between *about* 1700 and 1600 m.y., and that it continued for

a considerable time afterward (as suggested by the 'tail' of Nonacho site poles).

Site 1 in banded sandstone is very much more stable than the maroon sandstones (Fig. 3). Therefore its magnetization may be older and its pole is consistent with this view.

4. *Sequence of events.* We tentatively suggest the following sequence:

(*a*) Formation of the Archean basement prior to 2500 m.y.

(*b*) Faulting of the basement and deposition of Nonacho Group sediments some time between 2500–1700 m.y.

(*c*) Deep-burial, folding, and faulting at *approximately* 1800 m.y.

(*d*) Intrusion and magnetization of Sparrow dikes at about 1700 m.y.

(*e*) Magnetization of Nonacho sandstones by uplift following deep burial and mild heating, culminating between 1700 and 1600 m.y., but extending sporadically until 1400 m.y.

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